Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2020

Bachelor in Computer Applications

Course Title: **Microprocessor & Computer Architecture**

Code No: **CACS155** Semester: **II**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: *Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

2. Define CPU. Differentiate between Microprocessor and Microcontroller with example.

3. Define instruction cycle. Explain the opcode fetch machine cycle for MOV A, B with timing diagram. (Opcode: MOV A, B = 78h)

4. Write an ALP using 8085 to check number stored in memory location 8080h is either even or odd.

5. What is cache memory? Explain the elements of cache design.

6. Explain the organization of Microprogrammed Control Unit.

7. What is pipeline? Explain the four segment instruction pipeline.

8. Write short notes on (any two):a) Accumulatorb) 8085 Interruptsc) Structure of Hard Disk

Group C

Attempt any TWO questions[2x10=20]

9. Define instruction set. Classify the instructions available in 8085 with example.

10. Define the addressing mode. Explain the various instruction addressing modes with example.

11. Define micro-program? Describe symbolic micro-program for instruction FETCH routine. Explain the organization of micro-program sequencer for control memory with suitable diagram.